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DUNMAN HIGH SCHOOL

Preliminary Examination

Year 6

H1 BIOLOGY

Paper 1 Multiple Choice

8875/01

22 Sep 2011

1 hour

Additional Materials: Multiple Choice Answer Sheet

READ THESE INSTRUCTIONS FIRST.

Write in soft pencil.

Do not use staples, paper clips, highlighters, glue or correction fluid.

Write your name, civics group and index number on the Answer Sheet in the spaces provided unless this has been done for you.

There are **thirty** questions in this paper. Answer **all** questions. For each question there are four possible answers, **A, B, C** and **D**.

Choose the **one** you consider correct and record your choice in the **soft pencil** on separate Answer Sheet [**optical answer sheet (OTAS)**] provided.

Read the instructions on the Answer Sheet very carefully.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer. Any rough working should be done in this booklet.

Calculators may be used.

Do not open this booklet until you are told to do so.

FOR EXAMINER'S USE		
Paper 1		/30
Paper 2	A1	/15
	A2	/25
	B	/20
Grand Total		/90
Percentage		/%
Grade		

This document consists of **15** printed pages and **1** blank page.

Multiple Choice Questions (30 marks)

Answer **all** questions.

- 1** The electron micrograph below shows a plant cell.



Which of the following organelle(s) produces adenosine triphosphate?

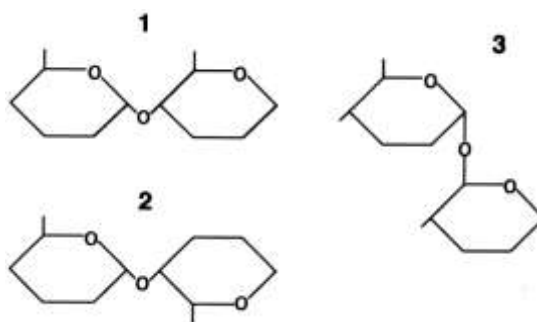
- A** D
- B** C and D
- C** D and E
- D** E and F

- 2 Which of the following is/are the **MOST** likely consequence/(s) for a cell lacking functional lysosomes?

- (i) The cell becomes crowded with undegraded wastes.
- (ii) The cell dies because its ATP-synthesizing mechanisms are missing.
- (iii) The cell dies from a lack of enzymes to catalyze metabolic reactions.
- (iv) The cell is unable to reproduce itself.
- (v) The cell is unable to grow to a mature size and always remains small.

- A (i) only
- B (i) and (v)
- C (ii) and (iv)
- D (iii) and (iv)

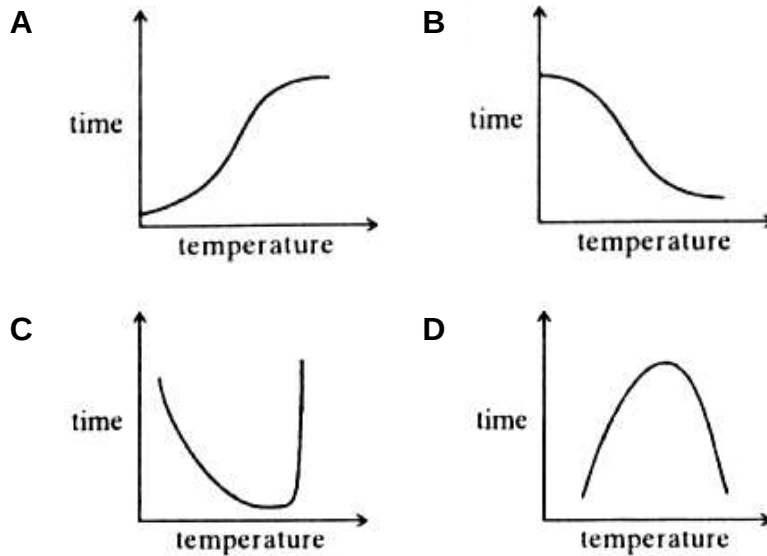
- 3 The diagrams show different types of bond in polysaccharides.



Which type of bond/(s) is/are found in amylose?

- A 1 only
 - B 2 only
 - C 1 and 3
 - D 2 and 3
- 4 How many different polypeptides, each consisting of r amino acids, can be made if the number of different amino acids available is n ?
- A n
 - B r^n
 - C nr
 - D n^r

- 5 In an investigation to determine the effect of temperature on the activity of an enzyme, the time taken for all the substrates to disappear from a standard solution was recorded.

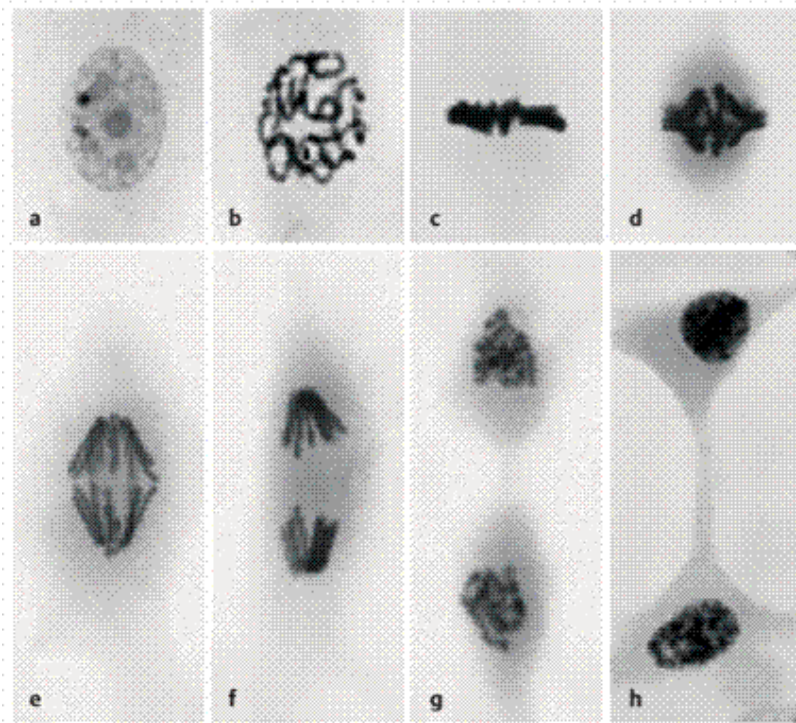


Which graph shows the result of this investigation?

- 6 Which of the following is the longest phase in the cell cycle of human liver cells?

- A prophase
- B anaphase
- C cytoplasmic cleavage
- D interphase

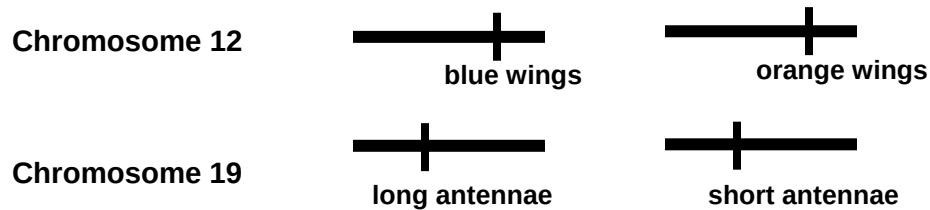
7 The micrographs below show some of the events that take place during mitosis.



Which of the following **CORRECTLY** describes the events?

	a	d	f	h
A	Prophase	Metaphase	Anaphase	Cytokinesis
B	Interphase	Anaphase	Telophase	Cytokinesis
C	Prophase	Anaphase	Telophase	Interphase
D	Interphase	Metaphase	Anaphase	Interphase

- 8 A certain (hypothetical) organism is diploid, has either blue or orange wings as the consequence of one of its genes, and has either long or short antennae as the result of a second gene, as shown below.



If an organism is heterozygous for both genes, what will be the possible chromosomes and alleles found in one of her eggs?

- A Two chromosomes 12 with blue and orange wings alleles.
 - B Chromosome 12 with an orange wings allele and chromosome 19 with long antennae allele.
 - C Chromosome 19 with both orange wings allele and short antennae allele due to crossing over.
 - D Chromosome 12 with blue and orange wings alleles and chromosome 19 with long and short antennae alleles.
- 9 Down's syndrome can exist in two forms.

Classic Down's syndrome affects all cells, while the mosaic form arises from the presence of two or more cell types in the body with differences in the chromosome number and structure.

How does the mosaic condition arise?

- A non-disjunction of chromosomes in mitosis in early foetal development
- B non-disjunction of chromosomes in mitosis during formation of ova
- C non-disjunction of chromosomes in mitosis during formation of sperm
- D translocation of chromosomes at maturation of ovum or the sperm

10 Nucleic acids have the following components.

1	2	3	4	5
phosphate	ribose	deoxyribose	purine	pyrimidine

Which components are linked by hydrogen bonds in DNA?

- A 1 and 3
- B 3 and 4
- C 2 and 5
- D 4 and 5

11 As a ribosome translocates along an mRNA molecule by one codon, which of the following occur(s)?

- (i) The tRNA that was in the A-site moves into the P-site.
- (ii) The tRNA that was in the P-site moves into the A-site.
- (iii) The tRNA that was in the E-site is released.
- (iv) A peptide bond is formed between the adjacent amino acids.

- A (i) and (ii)
- B (ii) and (iii)
- C (i), (iii) and (iv)
- D (ii), (iii) and (iv)

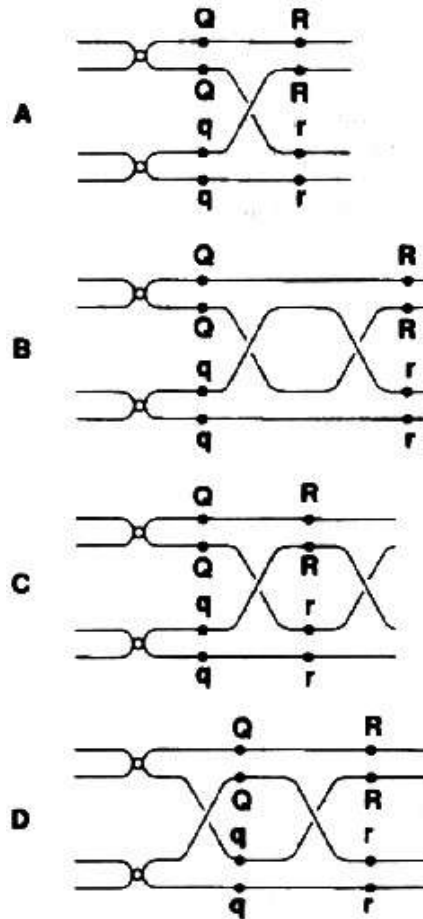
12 Two polypeptides were made using synthetic mRNA molecules as shown.

<u>Synthetic mRNA</u>	<u>Polypeptide produced</u>
AAAAAAAAAAAA	lysine-lysine-lysine-lysine
UUUAAAUUUAAA	phenylalanine-lysine-phenylalanine-lysine

What are the DNA codes for the amino acids phenylalanine and lysine?

	phenylalanine	lysine
A	AAA	TTT
B	AAA	UUU
C	GGG	CCC
D	TTT	GGG

- 13 Which diagram represents the situation which would result in the segregation of the alleles **Q** from **q**, and **R** from **r** in the first meiotic division?



- 14 In cats, black colour is caused by an X-linked allele and the other allele at this locus causes orange colour. The heterozygous is tortoiseshell.

What kinds of offspring would you expect from the cross of a black female and an orange male?

- A tortoiseshell female; tortoiseshell male
- B black female; orange male
- C orange female; orange male
- D tortoiseshell female; black male

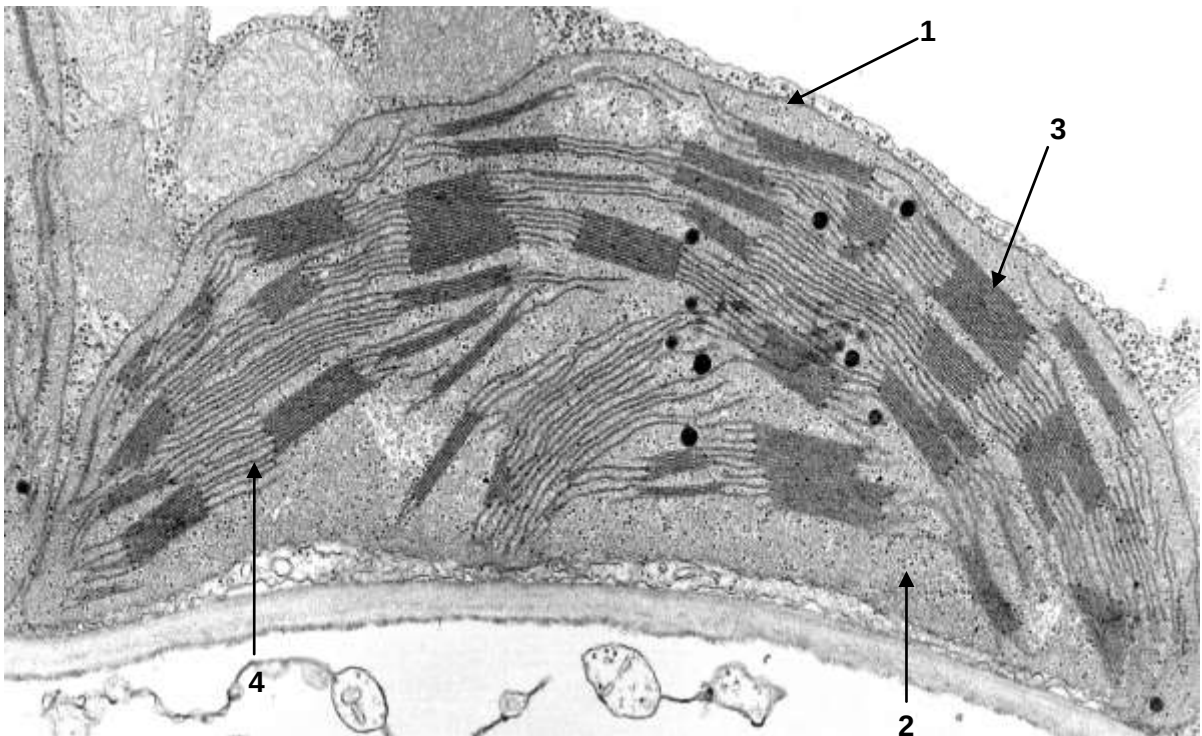
- 15 Day-old chicks can be difficult to sex, but their down feathers reveal a sex-linked dominant character called 'barred'. Female birds are **XY** and males are **XX**.

Which cross would ensure **CORRECT** sexing of the resultant chicks?

	female	male
A	barred	homozygous barred
B	barred	homozygous unbarred
C	unbarred	heterozygous barred
D	unbarred	homozygous barred

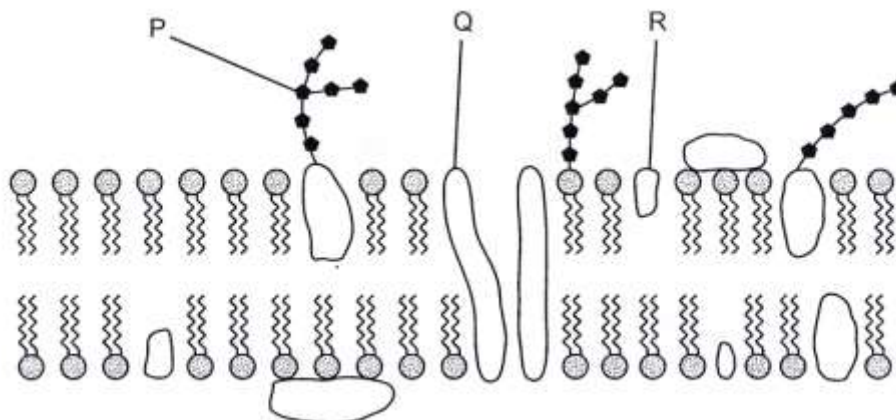
- 16 In animals, the amount of energy released from a given quantity of glucose is greater in aerobic respiration than in anaerobic respiration by approximately _____.
- A 32 times
B 19 times
C 9 times
D 3 times
- 17 Which of the following is the role of oxygen in the respiratory electron transfer system?
- A To accept hydrogen directly from organic acids.
B To accept hydrogen directly from reduced nicotinamide adenine dinucleotide.
C To act as the final acceptor of hydrogen at the end of the chain.
D To donate electrons to the respiratory chain.
- 18 The light reaction of photosynthesis does not include _____.
- A chemiosmosis
B pigment photoactivation
C NAD reduction
D electron transport

- 19 Which of the following is **CORRECTLY** associated with the function or characteristic of the labelled structure?



	help to maintain high H^+ concentration	made up of phospholipid	has stalked particles embedded in it	where ATP is synthesised
A	1 and 2	1, 2, 3 and 4	4	2
B	3 and 4	2 and 4	2 and 3	1 and 2
C	1 and 4	1, 2 and 3	1 and 4	2 and 4
D	3 and 4	1, 3 and 4	3 and 4	3 and 4

20 The diagram below shows part of a cell surface membrane.



What is the **CORRECT** function for each of the structures labelled?

	regulates membrane fluidity	forms hydrogen bonds with water to stabilise membrane	transports ions and large polar molecules
A	R	Q	P
B	P	Q	R
C	Q	R	P
D	R	P	Q

21 The allele that causes phenylketonuria (PKU) is harmful. What maintains the presence of this harmful allele in a population's gene pool?

- A** balanced polymorphism
- B** diploidy in organisms
- C** heterozygote advantage
- D** stabilising selection

22 In which of the following scenarios would allopatric speciation occur **MOST** quickly?

- A A population of cheetahs undergoing little or no evolutionary change.
- B A population of fish undergoing directional selection due to the introduction of a new predator.
- C A population of giraffes free to roam the entire nature reserve after the artificial removal of all predators.
- D A population of snails separated by a new fast-flowing river.

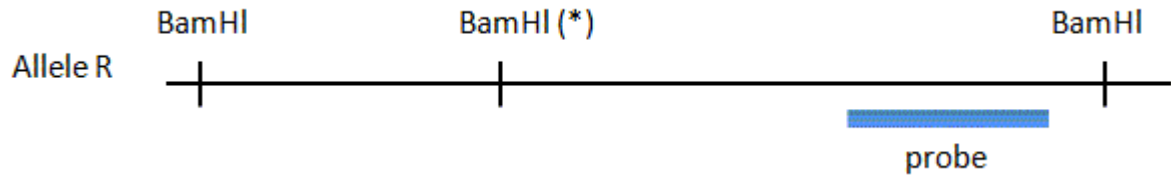
23 The table below shows the number of amino acid differences in the alpha haemoglobin chains between the crocodile and various organisms.

	lungfish	coelacanth	shark	goldfish
crocodile	85	78	77	70

Which organism is the **MOST** closely related to the crocodile?

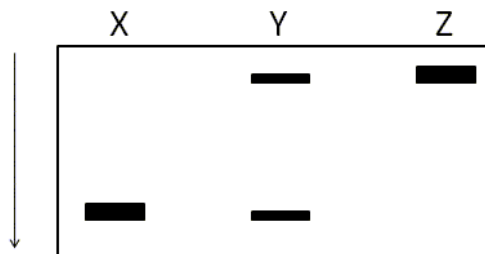
- A coelacanth
 - B goldfish
 - C lungfish
 - D shark
- 24 Which of the following characteristic is undesirable in cloning vectors used in genetic engineering?
- A ability to replicate in host
 - B have at least two genetic markers
 - C have at least two restriction sites of a particular restriction enzyme
 - D small in size

- 25 The diagram below shows the restrictions sites along allele R recognised by the restriction enzyme, *Bam*HI.



BamHI (*) indicates a mutation within the restriction site which give rise to recessive allele r.

DNA samples from three individuals, X, Y and Z, were subjected to restriction digestion by BamHI and separated by gel electrophoresis. The following results were obtained after probing.



Deduce the genotype of individual Z.

- A RR
- B Rr
- C rr
- D X^rY

26 Which uses of the information from the Human Genome Project are considered to be unethical?

- (i) an insurance company only giving better rates to people with genetic predispositions to fewer diseases
- (ii) genetic archaeologists identifying the earliest forms of genes to show evolutionary relationships
- (iii) cytologists developing tests for only some defective genes
- (iv) doctors only giving specific drugs to block the actions of faulty genes to carriers of those genes
- (v) genetic counselors giving specific lifestyle information only to people genetically predisposed to risks
- (vi) parents choosing embryos for implantation only after antenatal tests for acceptable genes

- A (i) and (iii)
- B (i) and (vi)
- C (ii) and (v)
- D (iii) and (iv)

27 Which of the following statement regarding stem cells is **FALSE**?

- A One potential side effect of any embryonic stem cell-based therapy is the formation of a tumour.
- B Stem cells are present within various organs of the adult body.
- C Stem cells can develop into a whole organism when implanted into the womb.
- D Stem cells can be grown and expanded indefinitely in culture under appropriate conditions.

- 28 Which feature of stem cells obtained from blood in the umbilical cord enables their use in the treatment of a variety of blood cancers?
- A They can differentiate into bone marrow cells.
 - B They can differentiate into any cellular component of blood.
 - C They can replace blood stem cells affected by cancer.
 - D They are totipotent.
- 29 Which genetic modification would **NOT** decrease the quantity of chemicals sprayed onto crop plants by farmers?
- A fungus resistance
 - B virus resistance
 - C insect resistance
 - D herbicide resistance
- 30 In terms of containment (prevention of genetic pollution of wild plant or non-GM crop populations), which of the following is an advantage of chloroplast transformation (introducing foreign genes into chloroplasts of host plants) over nuclear transformation?
- A Chloroplasts are surrounded by a double membrane.
 - B Chloroplasts are smaller than the nucleus.
 - C There are no chloroplasts in pollen of most plant species.
 - D There is no DNA in chloroplasts.

END OF PAPER

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